Researchers uncover similarities between human and AI learning

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By working to understand how new AI systems integrate flexible and incremental learning, researchers gained insights about the human brain.



PROVIDENCE, R.I. [Brown University] — New research found similarities in how humans and artificial intelligence integrate two types of learning, offering new insights about how people learn as well as how to develop more intuitive AI tools.

Led by Jake Russin, a postdoctoral research associate in computer science at Brown University, the study found by training an AI system that flexible and incremental learning modes interact similarly to working memory and long-term memory in humans.

"These results help explain why a human looks like a rule-based learner in some circumstances and an incremental learner in others," Russin said. "They also suggest something about what the newest AI systems have in common with the human brain."

Russin holds a joint appointment in the laboratories of Michael Frank, a professor of cogntive and psychological sciences and director of the Center for Computational Brain Science (https://ccbs.carney.brown.edu/) at Brown's Carney Institute for Brain Science, and Ellie Pavlick, an associate professor of computer science who leads the AI Research Institute on Interaction for AI Assistants (https://www.brown.edu/news/2025-07-29/aria-ai-institute-brown) at Brown. The study was published in the Proceedings of the National Academy of Sciences (https://www.pnas.org/doi/10.1073/pnas.2510270122).

Depending on the task, humans acquire new information in one of two ways. For some tasks, such as learning the rules of tic-tac-toe, "in-context" learning allows people to figure out the rules quickly after a few examples. In other instances, incremental learning builds on information to improve understanding over time — such as the slow, sustained practice involved in learning to play a song on the piano.

While researchers knew that humans and AI integrate both forms of learning, it wasn't clear how the two learning types work together. Over the course of the research team's ongoing collaboration (https://carney.brown.edu/news/2024-04-01/llms), Russin — whose work bridges machine learning and computational neuroscience — developed a theory that the dynamic might be similar to the interplay of human working memory and long-term memory.

To test this theory, Russin used "meta-learning"— a type of training that helps AI systems learn about the act of learning itself — to tease out key properties of the two learning types. The experiments revealed that the AI system's ability to perform in-context learning emerged after it meta-learned through multiple examples.

One experiment, adapted from an experiment in humans, tested for in-context learning by challenging the AI to recombine similar ideas to deal with new situations: if taught about a list of colors and a list of animals, could the AI correctly identify a combination of color and animal (e.g. a green giraffe) it had

not seen together previously? After the AI meta-learned by being challenged to 12,000 similar tasks, it gained the ability to successfully identify new combinations of colors and animals.

The results suggest that for both humans and AI, quicker, flexible in-context learning arises after a certain amount of incremental learning has taken place.

“At the first board game, it takes you a while to figure out how to play,” Pavlick said. “By the time you learn your hundredth board game, you can pick up the rules of play quickly, even if you’ve never seen that particular game before.”

The team also found trade-offs, including between learning retention and flexibility: Similar to humans, the harder it is for AI to correctly complete a task, the more likely it will remember how to perform it in the future. According to Frank, who has studied this paradox in humans, this is because errors cue the brain to update information stored in long-term memory, whereas error-free actions learned in context increase flexibility but don’t engage long-term memory in the same way.

For Frank, who specializes in building biologically inspired computational models to understand human learning and decision-making, the team’s work showed how analyzing strengths and weaknesses of different learning strategies in an artificial neural network can offer new insights about the human brain.

“Our results hold reliably across multiple tasks and bring together disparate aspects of human learning that neuroscientists hadn’t grouped together until now,” Frank said.

The work also suggests important considerations for developing intuitive and trustworthy AI tools, particularly in sensitive domains such as mental health.

“To have helpful and trustworthy AI assistants, human and AI cognition need to be aware of how each works and the extent that they are different and the same,” Pavlick said. “These findings are a great first step.”

The research was supported by the Office of Naval Research and the National Institute of General Medical Sciences Centers of Biomedical Research Excellence.

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